
Part VIII

Future Perspectives

Future Aspects: Climbing in the Olympics

23

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23.1 Competitive Sport Climbing

Climbing is currently experiencing a “boom” that has rapidly transformed it from a niche sport to a widely practiced popular sport [1]. The worldwide “boom” of the sub-discipline of bouldering and increasingly exciting competition formats in competitive sports has certainly played a decisive role in this development (Figs. 23.1 and 23.2) [2]. The great public interest is emphasized by the World Cup livestreams of all climbing disciplines, the participation in major sporting events such as the “World Games,” and the inclusion in the Olympic program. The IFSC (International Federation of Sport Climbing) World Cup circuit as well as major national and international competitions (world championships, Olympic qualifying competitions, etc.) has recently been attracting an increasingly large audience (Fig. 23.3). One of the main reasons for this is a new competi-

tion format adapted to the interests of worldwide (TV) viewers, in which two or more athletes no longer climb simultaneously but start one after the other. Optimized camera settings and specialized commentators provide additional excitement. Spectator records at the competition sites are regularly broken, and TV/ livestream ratings continue to increase.



Fig. 23.1 Competitive athlete in a boulder final 2020. (M. Kleesattel, Germany; Photo: Lutter)



Fig. 23.2 Jernej Kruder (left) and Adam Ondra (right) during the IFSC Boulder World Cup in Munich 2019. (Photo: Lutter)



Fig. 23.3 A large crowd of visitors during the qualification day of the IFSC Boulder World Cup in Munich 2019. (Photo: Lutter)

After the initial euphoria, the inclusion in the Olympic program caused a certain amount of uncertainty in the climbing community, especially among the athletes [1]. Due to the strict specifications of the organizing committee, a “climbing triathlon” was initially chosen as the competition format for Tokyo 2020, combining the previously separate competition formats of bouldering, lead and speed climbing (so-called Combined). For the 2024 Games in Paris, a separation of the sub-disciplines is planned. Based on the changed demands on the athletes, well-thought-out support of new talents is indispensable in order to avoid known climbing injuries and to prevent new injury patterns and overload reactions associated with the competition formats [3].

In addition, a trend that has been apparent for some time is likely to continue in the coming years; the average age of the top athletes is continuously decreasing, and both on real rock and in competition formats, most records are set by young or even very young athletes [1]. The inclusion in the Olympic program could reinforce this trend. Therefore, special attention must be paid to typical climbing injuries and overuse injuries in young athletes [4]. Injuries in the growth plates of the fingers and eating disorders (anorexia athletica) are the two most frequent and important entities among young athletes.

The authors therefore recommend the following steps and measures to ensure that the sport is safe and free of injuries in the future:

1. Structured, evidence-based, sport-specific medical supervision for all ambitious athletes
2. Supervision and adjustment of training and competition sites according to official recommendations and standards
3. Further development and consistent use of low-injury climbing holds and adapted route construction
4. Age-specific training and competition adjustment, especially for “Combined” athletes [1]

If these measures are considered and consistently implemented, it should also be possible to ensure largely injury-free training and competi-

tion conditions for competitive athletes in the future of climbing sports.

23.2 Anti-Doping

For athletes, trainers, and fans of the sport, the rapid rise of climbing as a competitive sport is an exciting development. However, the new Olympic status as well as the increasing professionalization comes with added responsibility. With the growing pressure to perform in competitive sports, the doping problem will inevitably become more important. In modern sport climbing, performance-enhancing substances can be used for both physical and mental performance enhancement. Competition climbing federations have adapted anti-doping rules or adopted regulations that had been standardized in other sports many years ago. So far, there is no corresponding scientific basis or evidence regarding the doping problem in climbing sport [2, 5]. This is in contrast to other new Olympic sports such as baseball or karate and reveals an urgent need to catch up. The state of knowledge about performance-enhancing substances in modern climbing is therefore still manageable and is largely based on anecdotal evidence and assumptions. Since a high strength-to-body weight ratio is crucial in climbing, low body weight has always been one of the decisive performance-limiting factors. Various substances such as appetite suppressants, diuretics, and other supplements have been used by climbers in the past. To prevent anorexic disorders in athletes, body weight or body mass index limits have been introduced in competition. However, severe sanctions have not yet been implemented. Instead, recommendations and warnings are issued to the relevant federations in the event of abnormalities. There is still little evidence of the use of stimulants to prevent fatigue, optimize training, or improve performance in competition. However, prohibited substances used in other sports also allow more intense and frequent training sessions in climbing, a shortened recovery period, and improved performance in competition.

Special importance is attached to anxiolytic and disinhibitory substances such as THC, which have positive psychological effects on the athletic performance of climbers and are unfortunately widespread in parts of the climbing community. The use of so-called low psychoactive cannabinoids (CBD oils, etc.) must also be viewed critically, even though these substances are not currently considered doping substances by the World Anti-Doping Organization (WADA) [2].

A further problem within the climbing sport is the partially insufficiently developed sports-medical supervision of (top) athletes. While internationally competing athletes from larger federations receive appropriate support from their national federations, athletes from smaller nations without sufficient funds sometimes do not receive sufficient sports-medical support at all (especially anti-doping).

The following measures are therefore recommended:

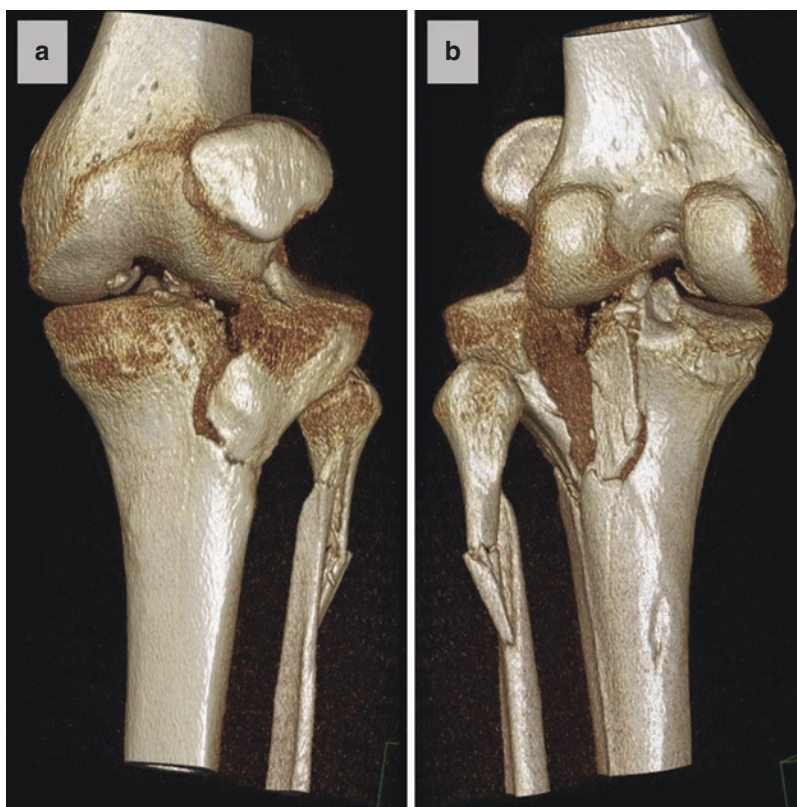
1. Increased awareness of sports ethics in climbing
2. Increased anti-doping education of fans, athletes, trainers, and officials
3. Sports-medical supervision for all competitive athletes
4. Cooperation between national sports organizations in anti-doping efforts
5. Increased monitoring and anti-doping controls (training and competition)

23.3 Newbie Syndrome

Due to the enormous popularity of bouldering, there has been a significant shift in the number of climbing sport injuries in the last 5 years [6]. Among these injuries, more severe injuries have been reported, with the emphasis on indoor bouldering. In contrast to older surveys of injury severity in climbing and bouldering, a slightly higher number of more severe climbing injuries are currently being observed. When analyzing this current trend, the change of the bouldering sport and the development of modern boulder-

ing gyms must be especially considered. With the emergence of bouldering competitions, the design of bouldering routes has changed. The attractiveness of the sport for athletes and spectators is improved by adapting the route construction, demanding more and more spectacular and acrobatic movements. Modern boulder problems are often a combination of classical climbing and bouldering and acrobatic elements as seen in parkour. This requires different skills than in the past, both from competitive athletes and from recreational athletes who often do similarly spectacular boulder problems. Most of the movements are dynamic and are often performed in a three-dimensional, overhanging wall structure. Special foot techniques such as so-called heel hooks are increasing frequently, which explains the increase of knee and hip injuries as a logical consequence of the changes in the sport. While the climbing or bouldering style and thus the athletic requirements change, another development can be observed parallel to this: a change in the athlete collective. Bouldering gyms have become popular meeting points and destinations for various events; now, company events and meetings, children's birthday parties, etc. are sometimes held in boulder gyms. Visiting a bouldering gym has also become an important leisure activity for many people. This leads to a new injury mechanism, which we recently called "newbie" syndrome [7]. Among the large number of absolute beginners, there are some physically unfit and coordinatively limited climbers who are interested in climbing. In such cases, even small falls can cause serious injuries, such as vertebral or luxation fractures (Fig. 23.4). Balance, body awareness, and control are also less developed in absolute beginners than in athletes who have been practicing the sport for a long time and who had started climbing before acrobatic elements became common. Therefore, the current standards of modern boulder gyms must be adapted for injury prevention. Structured training (correct falling and rolling, etc.) and an introduction to the sport and its hazards is of utmost importance for beginner athletes.

Fig. 23.4 Tibial plateau fracture (AO 41-B3, Schatzker II) with luxation of the patella and fracture of the proximal fibula (right knee joint) in a 36-year-old athlete after a fall during indoor bouldering despite sufficient floor mats. (a) Ventral view of the knee joint, (b) posterior view of the knee



23.3.1 Conclusion

Regarding the large stage upon which climbing sports can and will present itself in the future, high demands are placed on sports physicians. Not only must injury patterns and overuse damage be quickly recognized and treated, but the sports physicians also play an important role in the implementation of preventive measures and anti-doping work. Coaches, parents, and athletes rely on appropriate advice; only then can this inspiring sport continue to be valued and practiced by everyone as a spectacular, easily accessible, and health-promoting sport.

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